

SAT Math Mastery · Trick Pack 1G**SAT Trick Pack 1 — Part G — Answer Key & Worked Solutions**

Time	75 minutes: 55 min solving + 20 min self-check and error log
Topics	absolute-value bounds · scale chains · ordered-data extremes · similar solids · fraction equations · tangent slopes · parallel lines · fractional exponents · symmetry · circle loci · decay periods · radical systems
Name / Date	_____

1 · SKILLS & IDEAS

- S1.** Bound a negatively scaled absolute value by its maximum and turn “no solution” into a strict inequality.
- S2.** Chain a scale-copy enlargement, a drawing scale and a squared unit conversion in the right order.
- S3.** Maximise the mean of a subset of an ordered list by pushing every other value to its minimum (repeats allowed).
- S4.** Scale areas by k^2 and volumes by k^3 ; recover k from an area or volume ratio.
- S5.** Clear denominators with the LCM, distribute every minus sign, then evaluate and locate a derived expression.
- S6.** Find tangent slopes from an external point with the discriminant or the distance formula; use symmetry for the product.
- S7.** Match corresponding angles across a parallel segment and feed them into the angle sum.
- S8.** Rewrite a radical formula with fractional exponents and invert it without flipping the wrong fraction.
- S9.** A symmetric equation with exactly one solution has that solution on the axis of symmetry.
- S10.** Locate circle centres from tangency to the axes or from a centre confined to an axis; use chord half-lengths.
- S11.** Read the period of a decay factor from the exponent’s divisor and convert to a percent.
- S12.** Replace radicals by new variables and use sum, difference and product identities instead of solving.

2 · REQUIRED FACTS

- $|u| \geq 0$, so $c - d|u| \leq c$ when $d > 0$; the maximum occurs where $u = 0$.
- Lengths scale by k , areas by k^2 , volumes by k^3 . A linear conversion factor must be squared for area.
- If the values are positive integers and no distinctness is required, “smallest allowed” means 1 repeated.
- Dividing or multiplying an inequality by a negative number reverses it.
- The radius to a point of tangency is perpendicular to the tangent line; a perpendicular from the centre to a chord bisects the chord.
- A circle tangent to both axes in the first quadrant has centre (r, r) .
- A segment parallel to a side of a triangle cuts off corresponding (equal) angles with the other two sides.
- $a^{-p} = \frac{1}{a^p}$, $\sqrt[n]{a^m} = a^{m/n}$, $(a^{p/q})^{q/p} = a$.
- If $F(m - t) = F(m + t)$ for all t and $F(x) = k$ has exactly one solution, that solution is $x = m$.

3 · REQUIRED FORMULAS

$$\frac{V_2}{V_1} = \left(\frac{S_2}{S_1}\right)^{3/2}, \quad d(P, \ell) = \frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}}, \quad r^2 = d^2 + \left(\frac{c}{2}\right)^2$$

$$(a - b)^2 = a^2 + b^2 - 2ab, \quad (a + b)^2 = a^2 + b^2 + 2ab, \quad r_1 + r_2 = -\frac{b}{a}, \quad A(t) = A_0 g^{t/T}$$

4 · MINI THEORY SHEET

Extremes of absolute values. For $c - d|u|$ with $d > 0$, the largest value is c (at $u = 0$). If the other side is larger than c there is no solution; equality gives exactly one; smaller gives two. Divide by a negative number only after writing the inequality, and reverse it.

Ordered data. List the values in order, mark which positions are pinned (the median) and which are free; give each free position its least allowed value, subtract from the total, and divide by the size of the subset you are maximising.

Scale chains. Write the chain in the order it was applied and undo it in reverse. Convert units after finding the area, using the square of the linear factor.

Circles. Distances to the centre are all equal to r : equate squared distances to remove square roots. A chord, the perpendicular from the centre and the radius form a right triangle.

Decay periods. In $g^{t/T}$ the factor g is applied once every T time units; percent change = $g - 1$.

Speed shortcuts.

Situation	Shortcut
$c - d u = \text{expression}$, no solution	expression $> c$ (strict); reverse the inequality when dividing by a negative
copy enlarged, then scale	divide by the copy factor first; square every linear conversion for area
maximise a subset's mean	others at their minimum (repeats allowed), median at its value
area $\rightarrow$ volume (or back)	$k = \sqrt{\text{area ratio}}$ or $\sqrt[3]{\text{volume ratio}}$, then the other power
fractions in an equation	multiply by the LCM at once; rewrite each subtraction with brackets
tangents from a point level with the centre	slopes are $\pm m$; product = $-m^2$ with $\sin \theta = \frac{r}{d}$
parallel segment in a triangle	angle at D equals angle at B (same side of the transversal)
$P = a w^{-p/q}$	$w = \left(\frac{a}{P}\right)^{q/p}$: invert the base <i>and</i> the exponent
symmetric equation, one solution	the solution is the axis; evaluate the left side there
circle tangent to both axes	centre (r, r) ; two radii from a quadratic: sum = $-\frac{b}{a}$
$g^{t/T}$	the factor acts once per T units; not per year, not per $2T$
radicals in a system	set $a = \sqrt{x}$, $b = \sqrt{y}$; use $(a \mp b)^2$ to reach the target

5 · GUIDED EXAMPLES

Example 1 — Negative absolute value has a ceiling

For which values of k does $6 - 3|x - 2| = 2k$ have no solution?

The left side is at most 6 (when $x = 2$). No solution when $2k > 6$, so $k > 3$. If $k \leq 3$, solve $|x - 2| = \frac{6-2k}{3} \geq 0$: there are solutions.

Example 2 — Centre on an axis: equate squared distances

A circle has its centre on the x -axis and passes through $(1, 3)$ and $(5, 1)$. Find r^2 .

Centre $(h, 0)$: $(h - 1)^2 + 9 = (h - 5)^2 + 1 \Rightarrow -2h + 10 = -10h + 26 \Rightarrow h = 2$.

$r^2 = (2 - 1)^2 + 3^2 = 10$.

Example 3 — Ordered data: push the others to their minimum

Seven positive integers (repeats allowed) have sum 60 and median 8. Greatest possible mean of the three largest? Sorted $a_1 \leq \dots \leq a_7$, $a_4 = 8$. Make $a_1 = a_2 = a_3 = 1$. Then $a_5 + a_6 + a_7 = 60 - 3 - 8 = 49$, with each at least 8.

Mean = $\frac{49}{3}$.

6 · PRACTICE SET — 16 PROBLEMS · ALL SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1.

SAT CHALLENGE · DESMOS

For an integer k , the equation

$$9 - 3|2x - 8| = 1 - \frac{5k}{2}$$

has no real solution x . What is the greatest possible value of k ?

Worked solution

1. The expression $9 - 3|2x - 8|$ is largest when $2x - 8 = 0$, where it equals 9. So the left side is at most 9.
2. There is no solution exactly when the right side is greater than 9: $1 - \frac{5k}{2} > 9$.
3. $-\frac{5k}{2} > 8 \Rightarrow k < -\frac{16}{5}$ (dividing by $-\frac{5}{2}$ reverses the inequality).
4. The integers satisfying $k < -3.2$ are $\dots, -5, -4$. The greatest is -4 .
5. (Check: $k = -4$ gives right side $11 > 9$, no solution; $k = -3$ gives $8.5 \leq 9$, and $|2x - 8| = \frac{1}{6}$ has solutions.)

Answer: -4

2.

SAT CHALLENGE · CALCULATOR

A site plan was drawn so that 1 centimeter on the plan represents 40 meters of ground. A copier then enlarged the plan to 125% of its size. On the enlarged copy, the parcel appears as a rectangle measuring 7.5 centimeters by 5 centimeters. What is the actual area of the parcel, in hectares? (1 hectare = 10,000 square meters.)

(A) 384

(B) 3.84 ✓

(C) 6

(D) 38.4

Worked solution

1. Undo the enlargement first: original plan lengths are $\frac{7.5}{1.25} = 6$ cm and $\frac{5}{1.25} = 4$ cm.
2. Apply the drawing scale: $6 \cdot 40 = 240$ m and $4 \cdot 40 = 160$ m.
3. Area = $240 \cdot 160 = 38,400$ square meters.
4. Convert with the squared factor: $\frac{38,400}{10,000} = 3.84$ hectares.
5. (Skipping the first step gives $300 \cdot 200 = 60,000 \text{ m}^2 = 6 \text{ ha}$; mis-converting 38,400 gives 384 or 38.4.)

Answer: (B)

3.

SAT CHALLENGE · NO CALCULATOR NEEDED

Nine positive integers, not necessarily distinct, have a sum of 100 and a median of 9. What is the greatest possible value of the mean of the four largest of these integers?

(A) 25

(B) 20.25

(C) 21.75 ✓

(D) 23.75

Worked solution

1. Order them $a_1 \leq a_2 \leq \dots \leq a_9$. The median is $a_5 = 9$, and $a_6, \dots, a_9 \geq 9$.
2. To make the top four as large as possible, make $a_1, \dots, a_4$ as small as possible. Repeats are allowed, so each can equal 1.
3. Then $a_6 + a_7 + a_8 + a_9 = 100 - (1 + 1 + 1 + 1) - 9 = 87$ (for example 9, 9, 9, 60, which respects the ordering).
4. Mean = $\frac{87}{4} = 21.75$.
5. (Forcing distinct values 1, 2, 3, 4 would give $\frac{81}{4} = 20.25$.)

Answer: (C)

4.

SAT CHALLENGE · CALCULATOR

Cylinder P has radius 2 centimeters and height 6 centimeters. Cylinder Q is similar to cylinder P and has a total surface area of 200π square centimeters. What is the volume, in cubic centimeters, of cylinder Q ?

(A) 375π ✓(B) 600π (C) 60π (D) 150π **Worked solution**

1. Total surface area of P : $2\pi r(r + h) = 2\pi(2)(8) = 32\pi$. Volume of P : $\pi r^2 h = 24\pi$.
2. Area ratio = $\frac{200\pi}{32\pi} = \frac{25}{4}$, so the linear factor is $k = \sqrt{\frac{25}{4}} = \frac{5}{2}$.
3. Volume ratio = $k^3 = \frac{125}{8}$.
4. $V_Q = 24\pi \cdot \frac{125}{8} = 375\pi$.
5. (Check: Q has radius 5, height 15; surface area $2\pi(5)(20) = 200\pi$, volume $\pi(25)(15) = 375\pi$.)

Answer: (A)

5.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$\frac{3x+2}{4} - \frac{x-7}{6} = \frac{5x-1}{12} + 2$$

If x is the solution to the equation above, which of the following intervals contains the value of $x^2 + 4x - 1$?

(A) Between 10 and 50

(B) Greater than 50

(C) Between 0 and 5

(D) Between 5 and 10 ✓**Worked solution**

1. Multiply every term by the LCM 12: $3(3x+2) - 2(x-7) = (5x-1) + 24$.

2. Distribute (the minus sign applies to both terms of $x-7$): $9x+6-2x+14 = 5x+23$, so $7x+20 = 5x+23$.

3. $2x = 3 \Rightarrow x = \frac{3}{2}$.

4. $x^2 + 4x - 1 = \frac{9}{4} + 6 - 1 = \frac{29}{4} = 7.25$, which lies between 5 and 10.

5. (A sign slip, $-2x-14$, gives $x = \frac{31}{2}$ and a value above 50.)

Answer: (D)

6.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the xy -plane, the circle with equation $(x-7)^2 + (y+2)^2 = 9$ has two distinct tangent lines that pass through the point $(2, -2)$. What is the product of the slopes of these two lines?

Worked solution

1. The point $(2, -2)$ is level with the centre $(7, -2)$, so the situation is symmetric about the horizontal line $y = -2$: the two tangent slopes are m and $-m$, and the product is $-m^2$.

2. Distance from the point to the centre: $d = 5$; radius $r = 3$. The tangent makes angle θ with the line to the centre where $\sin \theta = \frac{3}{5}$, so $\tan \theta = \frac{3}{4}$ and $m^2 = \frac{9}{16}$.

3. Check by the discriminant: substituting $y = -2 + m(x-2)$ gives $(1+m^2)x^2 - (14+4m^2)x + (40+4m^2) = 0$; its discriminant is $36 - 64m^2$, which is 0 for $m^2 = \frac{9}{16}$.

4. Product $= -\frac{9}{16}$.

Answer: $-9/16$

7.

SAT CHALLENGE · NO CALCULATOR NEEDED

In triangle ABC , point D lies on $\overline{AB}$ and point E lies on $\overline{AC}$ so that $\overline{DE}$ is parallel to $\overline{BC}$. The measure of $\angle ADE$ is $(4x-10)^\circ$, the measure of $\angle ABC$ is $(2x+30)^\circ$, and the measure of $\angle ACB$ is $(x+25)^\circ$. What is the measure, in degrees, of $\angle BAC$?

(A) 115

(B) 45

(C) 65 ✓

(D) 70

Worked solution

1. Since $\overline{DE} \parallel \overline{BC}$, with transversal $\overline{AB}$, the corresponding angles are equal: $\angle ADE = \angle ABC$.

2. $4x - 10 = 2x + 30 \Rightarrow x = 20$.

3. Then $\angle ABC = 70^\circ$ and $\angle ACB = 20 + 25 = 45^\circ$.

4. Angle sum in triangle ABC : $\angle BAC = 180 - 70 - 45 = 65^\circ$.

5. (Pairing $\angle ADE$ with $\angle ACB$ instead would give $x = \frac{35}{3}$ and a wrong triangle.)

Answer: (C)

8.

SAT CHALLENGE · CALCULATOR

$$P = \frac{a}{\sqrt[5]{w^2}}$$

The formula above relates the positive quantities P and w , where a is a positive constant. Which of the following expresses w in terms of P and a ?

(A) $w = \left(\frac{P}{a}\right)^{2/5}$

(B) $w = \left(\frac{a}{P}\right)^{5/2}$ ✓

(C) $w = \left(\frac{P}{a}\right)^{5/2}$

(D) $w = \left(\frac{a}{P}\right)^{2/5}$

Worked solution

1. Write the root as a power: $\sqrt[5]{w^2} = w^{2/5}$, so $P = a w^{-2/5}$.

2. Multiply by $w^{2/5}$ and divide by P : $w^{2/5} = \frac{a}{P}$.

3. Raise both sides to the reciprocal power $\frac{5}{2}$: $w = \left(\frac{a}{P}\right)^{5/2}$.

4. (Numerical check: $a = 7$, $w = 3.2$ gives $P = 7 \cdot 3.2^{-0.4}$, and $\left(\frac{a}{P}\right)^{5/2} = 3.2$.)

Answer: (B)

9.

SAT CHALLENGE · DESMOS

For a constant k , the equation

$$2\sqrt{x+1} + 2\sqrt{15-x} = k$$

has exactly one real solution. What is the value of k ?

(A) 16

(B) $2 + 2\sqrt{15}$

(C) 8

(D) $8\sqrt{2}$ ✓

Worked solution

1. The domain is $-1 \leq x \leq 15$. The two expressions under the radicals add to 16 and are symmetric about $x = 7$ ($x+1$ and $15-x$ swap when $x \rightarrow 14-x$), so if x is a solution then so is $14-x$.

2. Exactly one solution forces $x = 14-x$, i.e. $x = 7$.

3. $k = 2\sqrt{8} + 2\sqrt{8} = 4\sqrt{8} = 8\sqrt{2}$.

4. (Check: the left side rises from 8 at $x = -1$ to its maximum $8\sqrt{2} \approx 11.3$ at $x = 7$, then falls back to 8. Only the peak value is reached once. Other options give 0 or 2 solutions.)

Answer: (D)

10.

SAT CHALLENGE · NO CALCULATOR NEEDED

A rectangular banner has width w and length $1.8w$, where $w > 0$. The banner is covered exactly, with no gaps or overlaps, by 4 identical square tiles. The side length of each tile is $kw\sqrt{5}$, where k is a positive constant. What is the value of k ?

Worked solution

1. Banner area: $w \cdot 1.8w = 1.8w^2$.

2. Four tiles: $4(kw\sqrt{5})^2 = 4 \cdot 5k^2w^2 = 20k^2w^2$.

3. Set equal and cancel w^2 (the radical squares away): $20k^2 = 1.8 \Rightarrow k^2 = 0.09$.

4. $k > 0$, so $k = 0.3$.

Answer: 0.3

11.

SAT CHALLENGE · CALCULATOR

In the xy -plane, a circle lies in the first quadrant and is tangent to both the x -axis and the y -axis. The line $3x + 4y = 20$ intersects the circle at two points 8 units apart. What is the sum of all possible values of the radius of the circle?

Worked solution

1. Tangent to both axes in the first quadrant: centre (r, r) and radius r .
2. Distance from the centre to the line: $d = \frac{|3r + 4r - 20|}{5} = \frac{|7r - 20|}{5}$.
3. The perpendicular from the centre bisects the chord, so $d^2 + 4^2 = r^2$: $\frac{(7r - 20)^2}{25} + 16 = r^2$.
4. $49r^2 - 280r + 400 + 400 = 25r^2 \Rightarrow 24r^2 - 280r + 800 = 0 \Rightarrow 3r^2 - 35r + 100 = 0$.
5. Roots $r = 5$ and $r = \frac{20}{3}$, both positive and both valid (check $d = 3$ and $d = \frac{16}{3}$). Their sum is $\frac{35}{3}$ (Vieta: $-\frac{-35}{3}$).

Answer: 35/3

12.

SAT CHALLENGE · NO CALCULATOR NEEDED

The value V , in dollars, of a piece of lab equipment t years after it was purchased is modeled by

$$V(t) = 2400(0.64)^{t/2.5}.$$

Which of the following is the best interpretation of the model?

- (A) It falls 36% every 2.5 years. ✓ (B) It falls 36% every 5 years.
 (C) It falls 64% every 2.5 years. (D) It falls 2.5% every year.

Worked solution

1. The exponent is $\frac{t}{2.5}$, so it increases by 1 each time t increases by 2.5 years.
2. Each increase of 1 in the exponent multiplies V by 0.64. So every 2.5 years the value is multiplied by 0.64.
3. A factor of 0.64 is a decrease of $1 - 0.64 = 0.36 = 36\%$.
4. So the value decreases by 36% every 2.5 years.
5. (Over 5 years the factor is $0.64^2 \approx 0.41$, a 59% drop, not 36%.)

Answer: (A)

13.

SAT CHALLENGE · CALCULATOR

Rectangular box P has dimensions 3 inches by 4 inches by 6 inches. Box Q is similar to box P and has a volume of 1125 cubic inches. What is the total surface area, in square inches, of box Q ?

Worked solution

1. Volume of P : $3 \cdot 4 \cdot 6 = 72$. Surface area of P : $2(12 + 18 + 24) = 108$.
2. Volume ratio $\frac{1125}{72} = \frac{125}{8}$, so the linear factor is $k = \sqrt[3]{\frac{125}{8}} = \frac{5}{2}$.
3. Areas scale by $k^2 = \frac{25}{4}$: $108 \cdot \frac{25}{4} = 675$.
4. (Check: Q is $7.5 \times 10 \times 15$, volume 1125, surface area $2(75 + 112.5 + 150) = 675$.)

Answer: 675

14.

SAT CHALLENGE · NO CALCULATOR NEEDED

Positive real numbers x and y satisfy

$$\sqrt{x} - \sqrt{y} = 2 \quad \text{and} \quad x + y = 26.$$

What is the value of xy ?**Worked solution**

1. Let $a = \sqrt{x}$ and $b = \sqrt{y}$. Then $a - b = 2$ and $a^2 + b^2 = 26$.
2. Square the first: $a^2 - 2ab + b^2 = 4$, so $26 - 2ab = 4$ and $ab = 11$.
3. $xy = a^2b^2 = (ab)^2 = 121$.
4. (Check: $a = 1 + 2\sqrt{3}$, $b = 2\sqrt{3} - 1$ gives $a - b = 2$, $a^2 + b^2 = 26$, $ab = 12 - 1 = 11$.)

Answer: 121

15.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the xy -plane, the center of a circle lies on the y -axis. The circle passes through the points $(6, 1)$ and $(-2, 5)$. Which of the following is an equation of the circle?

(A) $x^2 + (y + 1)^2 = 44$

(B) $x^2 + (y + 1)^2 = 40$ ✓

(C) $x^2 + (y - 1)^2 = 40$

(D) $x^2 + (y + 1)^2 = 20$

Worked solution

1. The centre is $(0, k)$. Equal distances to the two points: $6^2 + (1 - k)^2 = (-2)^2 + (5 - k)^2$.
2. Expand: $36 + 1 - 2k + k^2 = 4 + 25 - 10k + k^2$, so $37 - 2k = 29 - 10k$.
3. $8k = -8 \Rightarrow k = -1$. Centre $(0, -1)$.
4. $r^2 = 6^2 + (1 - (-1))^2 = 36 + 4 = 40$. Check with the other point: $(-2)^2 + (5 + 1)^2 = 4 + 36 = 40$.
5. Equation: $x^2 + (y + 1)^2 = 40$.

Answer: (B)

16.

SAT CHALLENGE · CALCULATOR

For all real numbers x ,

$$8^x \cdot 4^{x/3} \cdot (\sqrt{2})^x = b^x,$$

where b is a positive constant. Which of the following is equal to 2?

(A) b^{25}

(B) $b^{-6/25}$

(C) $b^{6/25}$ ✓

(D) $b^{25/6}$

Worked solution

1. Write each factor as a power of 2: $8^x = 2^{3x}$, $4^{x/3} = 2^{2x/3}$, $(\sqrt{2})^x = 2^{x/2}$.
2. Multiply (add exponents): $2^{x(3 + \frac{2}{3} + \frac{1}{2})} = 2^{\frac{25}{6}x}$.
3. So $b^x = (2^{25/6})^x$ for all x , which gives $b = 2^{25/6}$.
4. Raise both sides to the reciprocal power $\frac{6}{25}$: $b^{6/25} = 2$.
5. (Evaluating at $x = 1$ gives b directly: $8 \cdot 4^{1/3} \cdot \sqrt{2}$.)

Answer: (C)